REALISTIC CONTEXT SYNTHESIS : LLM-GUIDED SLOT EDITING AND PHONOLOGICAL SPELLING AUGMENTATION FOR CODE-SWITCHED ASR

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ABSTRACT

Code-switched (CS) terms such as personal names, products and organizations often carry the key information of a VietnameseEnglish utterance, yet ASR systems trained on monolingual speech routinely fail to recognize them. Collecting natural CS speech to close this gap is costly, so such data remains scarce, while text-to-speech augmentation is only as useful as the CS text it renders. We present RCS, a text-side augmentation pipeline that preserves real Vietnamese transcripts as conversational frames and changes only where and how English enters them. An LLM proposes in-situ switch slots but never chooses the English words; slots are grounded in a curated catalog, allowing many controlled text variants and easy insertion of new terms. Before full-utterance TTS, embedded terms are optionally respelled into Vietnamese-adapted forms to model L1-conditioned pronunciation variation while labels remain canonical. RCS text is substantially closer to real CS speech than LLM rewriting. With 80k synthetic utterances, Whisper CS term recall rises from 29.2% to 45.3% and WER falls by 22% relative; gains also transfer to a Conformer-CTC system and an unseen medical domain.

***Index Terms*—** code-switching, speech recognition, data augmentation, text-to-speech, large language models

1. INTRODUCTION

Code-switching (CS) remains difficult for automatic speech recognition (ASR), especially when lexically sparse English names, products, and technical terms are embedded in Vietnamese speech [1, 2]. These terms are often the most informative words in an utterance, so errors on them are disproportionately costly [3]. Natural Vietnamese–English CS corpora are limited and domain-specific [4], making augmentation attractive.

Audio-side augmentation splices or concatenates monolingual speech [5, 6], preserving real acoustics but often creating unnatural boundaries. Text-side approaches instead generate CS text and synthesize it with TTS [7–9]. These text-side methods have evolved from rule-based substitution, which replaces translations or part-of-speech-guided spans under the equivalence constraint [10], to neural generation with GANs [7], and most recently to LLM-driven generation [8], which produces diverse code-switched transcripts at scale by prompting multilingual models with sentences in two languages [11] or by constrained decoding that enforces linguistic constraints [2, 12]; LLMs have also been used to augment textual data and rescore ASR hypotheses [13]. Yet two limitations remain. First, full generation or rewriting replaces the surrounding conversational register and lets the LLM choose an uncontrolled embedded vocabulary; LLM output also tends toward written-style text that loses the colloquial patterns of spontaneous speech.

Second, canonical English TTS ignores the Vietnamese-adapted pronunciations that bilingual speakers routinely produce. Embedded English items undergo L1-conditioned phonological transfer, adapting to Vietnamese phonology so their pronunciation deviates from the dictionary form and makes them behave like rare or out-of-vocabulary words. A natural remedy is spelling augmentation, which encodes these variants as phonetically plausible alternative spellings. Prior work shows that perturbing reference transcriptions with similar-sounding spellings improves rare-word robustness [14], that LLM-driven phonetic respelling broadens lexical and phonetic coverage [15], and that shared phonetic structure eases cross-lingual transfer [16].

We propose RCS (**R**ealistic **C**ontext **S**ynthesis), which preserves each real Vietnamese transcript and edits only selected switch positions. Our contributions are twofold: (1) *catalog-grounded text variants*: an LLM proposes plausible replacement/insertion slots, which are grounded to a curated catalog so a single transcript yields many controlled CS variants without asking the LLM to invent the English term; and (2) *respelling augmentation*: embedded terms are stochastically rendered with Vietnamese-adapted spellings before TTS, while training labels remain canonical. We evaluate text naturalness, two ASR model families, out-of-domain transfer, training strategies, respelling rate, decoder design, and scaling.

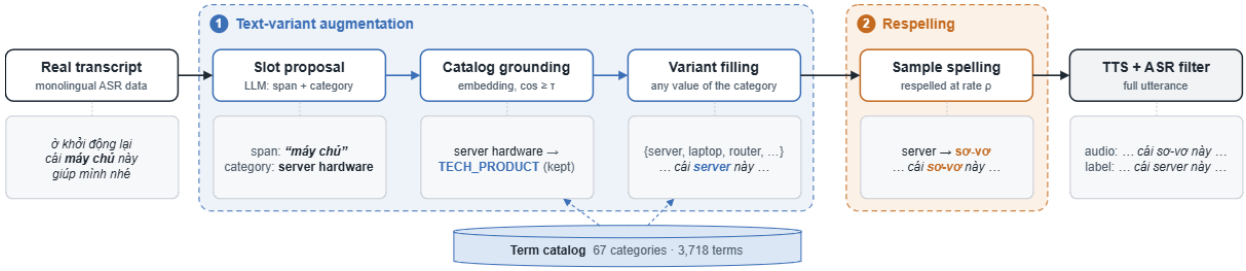


Fig. 1. Overview of RCS. ① Slot proposal and catalog grounding turn one real transcript into many controlled CS variants; ② embedded terms are respelled at rate ρ before full-utterance TTS, while ASR labels keep canonical spellings.

2. METHOD

2.1. In-situ slot proposal and grounding

Let $\mathcal{D}_{\text{mono}} = \{(x_i, T_i)\}_{i=1}^{N}$ be monolingual Vietnamese speech with transcript $T = (w_1, \ldots, w_n)$, and let $\mathcal{E} = \{(c, d_c, \mathcal{V}_c)\}$ be a catalog whose category c has description d_c and English values $\mathcal{V}_c$. Given T, an LLM proposes $p_j = (o_j, a_j, t_j, \delta_j)$, where $o_j \in \{\text{REP}, \text{INS}\}$, a_j is a replacement span or insertion anchor, and (t_j, δ_j) describes the semantic type. The LLM never writes the final CS sentence and never selects the English term.

Each proposal is mapped to the fixed catalog by

$$\hat{c}_j = \arg\max_{c \in \mathcal{E}} \cos(\phi(t_j \| \delta_j), \phi(c \| d_c \| \mathcal{V}_c)), \tag{1}$$

and retained only above similarity threshold $\tau = 0.78$. For admitted slots $\mathcal{A}(T)$, we sample non-overlapping subsets and catalog values to obtain

$$\mathcal{V}(T) = \{T \oplus \{(a_j, e_j)\}_{j \in S} : S \subseteq \mathcal{A}(T),\ e_j \in \mathcal{V}_{\hat{c}_j}\}. \tag{2}$$

Everything outside edited positions is copied from the real transcript, retaining discourse particles, disfluencies, and fragmentary conversational syntax. Cached slots can host newly added catalog terms without another LLM proposal pass.

2.2. Respelling and rendering

For each catalog value e, an LLM generates once, offline, Vietnamese-adapted spellings $\mathcal{S}(e)$ (e.g., *server*→{s-v, x-v}). For a fraction ρ of synthetic utterances, all embedded terms in the synthesis string are replaced by sampled respellings; otherwise canonical spellings are used. The ASR label always retains canonical terms. Thus $\rho = 0$ is canonical TTS and $\rho = 1$ always uses respelled synthesis.

Each synthesis string is rendered end-to-end with OmniVoice [17], using the original recording as speaker prompt; no waveform segment is edited. A lightweight ASR check removes gross synthesis failures. The retained synthetic pairs are mixed with real monolingual speech for training.

3. EXPERIMENTAL SETUP

3.1. Data and text baselines

Real training speech is 200k Vietnamese utterances from GigaSpeech2 [18]. RCS uses a general-domain catalog of 67 categories and 3,718 terms. From 50.6k source transcripts we obtain at least 5.2M possible variants and sample an 83.2k pool; 98.9% of terms have a non-canonical respelling. Unless stated otherwise, experiments use 80k synthetic utterances with $\rho = 0.2$. Evaluation uses Vietnamese-only VietnamTest, general-domain GigaSpeech2 CS, and medical-domain ViMedCSS [4]. For text quality, we compare against LLM generation from scratch and a transformation prompt following [2], matched to 10k outputs and evaluated against real GigaSpeech2 CS windows.

3.2. ASR systems and metrics

Whisper [19] is Whisper-large-v3-turbo fully fine-tuned with AdamW. **Conformer-CTC** [20, 21] is trained from scratch on Vietnamese onset–rime–tone units and decoded with a 4-gram LM plus lexicon. We report WER, CER, and CS term recall (TR): the fraction of reference CS terms exactly recovered. We compare **Mono**, sequential **Mono→CS**, and **Mix** training. Text quality uses PPL gap, CMI gap, MAUVE [22], C2ST [23], switched function-word rate, Vendi [24], and Self-BLEU.

4. RESULTS AND DISCUSSION

4.1. Text naturalness

Table 2 shows that RCS is closest to real CS speech across naturalness and diversity metrics. Its PPL gap is 0.07 versus 0.17 for transformation and 0.68 for generation from scratch, while its MAUVE is 0.36 versus 0.21 and 0.03. Transformation switches English function words far too often (26.0% versus 0.3% in real speech), whereas RCS retains the real Vietnamese frame and nearly matches real-text diversity (Vendi 4.39 versus 4.43). Catalog expansion also yields more than 100 variants per LLM proposal call without asking the generator to rewrite the sentence.

Table 1. ASR results (%). GigaSpeech2 CS: general-domain CS; ViMedCSS: unseen medical-domain CS; VietnamTest: Vietnamese only. Rows give the number of added RCS utterances and the respelling rate ρ. *Measured with 50k RCS utterances.

	GigaSpeech2 CS			ViMedCSS			VietnamTest*	
Training	WER↓	CER↓	TR↑	WER↓	CER↓	TR↑	WER↓	CER↓
Whisper (fine-tuned)								
Mono (200k)	20.76	10.72	29.24	33.15	22.70	11.76	25.80	15.62
+RCS 80k no augment spelling	18.42	9.41	39.18	32.57	21.91	14.62	23.54	14.08
+RCS 80k augment spelling	**16.10**	**7.90**	**45.30**	**31.95**	**21.24**	**17.69**	**22.70**	**13.50**
Conformer-CTC + LM (from scratch)								
Mono (200k)	26.00	15.58	7.46	39.82	27.41	4.93	28.74	17.56
+RCS 80k + augment spelling	**21.50**	**10.80**	**18.80**	**35.96**	**23.68**	**11.72**	**27.61**	**16.74**

Table 2. Generated text (10k utterances per method) versus length-matched real CS windows. †: CS spans masked. Target values follow the real CS speech row (C2ST 0.50, Vendi 4.43, ΔSelf-BLEU 0). Yield: CS sentences per LLM call (‡: lower bound).

	Naturalness					Diversity		Cost
Method	PPL gap† ↓	CMI gap↓	MAUVE↑	C2ST†	Func. sw. (%)	Vendi	ΔSelf-BLEU†	Yield↑
Real CS speech	0	0	1.00	0.50	0.3	4.43	0	–
LLM from scratch	0.68	1.22	0.03	0.98	3.7	3.69	+10.7	1
Transformation prompt [2]	0.17	7.49	0.21	0.77	26.0	4.17	+7.6	1
RCS	**0.07**	**1.13**	**0.36**	**0.73**	**0.2**	**4.39**	**−0.9**	**103**‡

Table 3. Training and evaluation data. CMI is averaged over all utterances (CS-only subset in parentheses); M/I are code-mixing indices; SP is switch points per utterance.

Set	Utts	Hrs	EN%	CMI	M	I	SP	Terms
Training								
GigaSpeech2 VN	200k	161.0	0	0	0	0	0	0
RCS	83.2k	55.6	15.5	18.6 (18.7)	.35	.20	1.63	3,718
Evaluation								
VietnamTest	66.6k	51.5	0	0	0	0	0	0
GigaSpeech2 CS	1.5k	3.9	5.5	6.0 (6.3)	.12	.08	2.94	1,217
ViMedCSS	1.6k	3.4	5.9	7.1 (7.4)	.13	.09	2.44	942

Table 4. Training strategy with Whisper (%). VN: VietnamTest; CS: GigaSpeech2 CS.

Strategy	VN WER↓	CS WER↓	CS TR↑
Mono	25.8	20.76	29.24
Mono→CS	28.5	17.10	42.00
Mix	**22.7**	**16.10**	**45.30**

Table 5. Respelling-rate ablation with Whisper on GigaSpeech2 CS (%). Each setting adds 10k RCS utterances.

ρ	WER↓	CER↓	TR↑
0.0	19.0	9.8	37.62
0.2	**17.4**	**9.0**	**39.00**
0.5	18.0	9.3	36.40
1.0	19.0	9.7	32.00

4.2. Main ASR results and transfer

Table 1 reports results across all three test sets. On GigaSpeech2 CS, RCS with spelling augmentation lowers Whisper WER from 20.76% to 16.10% (22.4% relative) and raises term recall from 29.24% to 45.30%, while spelling-free synthesis already yields strong gains (18.42% WER, 39.18% recall). Gains persist without multilingual pre-training: Conformer-CTC WER falls from 26.00% to 21.50%, while term recall increases from 7.46% to 18.80% (2.52×). Out-of-domain transfer to the unseen ViMedCSS set is also clear, with term recall rising from 11.76% to 17.69% (+50.4% relative). Improvements do not trade off against Vietnamese recognition: VietnamTest WER drops from 25.80% to 22.70% and CER from 15.62% to 13.50%.

4.3. Training and respelling

Table 4 follows the trend observed in prior CS training studies [2]: sequential adaptation to CS improves CS recognition but degrades the matrix language, whereas mixed training achieves the best CS result while preserving Vietnamese. Respelling behaves as augmentation rather than replacement (Table 5). Canonical-only synthesis ($\rho = 0$) already improves term recall, a moderate mixture ($\rho = 0.2$) is best on all metrics, and always respelling ($\rho = 1$) reduces term recall to 32.0%. Exposure to both canonical and Vietnamese-adapted

Table 6. Conformer-CTC decoder analysis on GigaSpeech2 CS (%). "Resp." adds Vietnamese-adapted CS pronunciations to the lexicon; Multi LM includes CS text.

Acoustic	LM	Lexicon	WER↓	CER↓	TR↑
Mono	Mono	Base	26.00	15.58	7.46
RCS	Mono	Base	21.50	10.90	15.20
RCS	Mono	+Resp.	21.50	10.80	18.80
RCS	Multi	+Resp.			

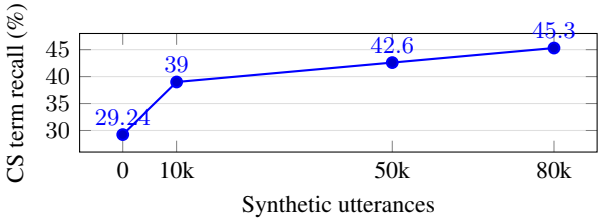


Fig. 2. Scaling RCS with fixed respelling rate $\rho = 0.2$. All synthetic sets are drawn from the same monolingual source corpus.

realizations is therefore preferable to either extreme.

4.4. Decoder analysis and scaling

The from-scratch Conformer separates acoustic and decoding effects (Table 6). RCS acoustic training more than doubles term recall even with a monolingual LM and base lexicon (7.46%→15.20%). Adding respelled lexical pronunciations increases recall to 18.80% with almost unchanged WER, indicating that pronunciation-aware decoding specifically recovers embedded terms. A CS-aware LM further raises recall to 20.40%. Finally, Fig. 2 fixes $\rho = 0.2$ and varies only synthetic set size: term recall grows from 29.24% without synthetic data to 39.00% at 10k, 42.60% at 50k, and 45.30% at 80k, showing continued benefit from catalog-driven variants without additional real speech.

5. CONCLUSION

We presented RCS, which preserves real Vietnamese conversational frames while grounding switch slots to a controllable English catalog and augmenting embedded-term pronunciation through Vietnamese-adapted respellings. The generated text better matches real CS speech than LLM rewriting, improves both pretrained and from-scratch ASR, transfers to an unseen medical domain, and scales with additional catalog-driven variants. On Whisper, 80k RCS utterances reduce general-domain CS WER by 22.4% relative and raise CS term recall from 29.24% to 45.30% without sacrificing Vietnamese recognition.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study uses public speech corpora and generated speech for ASR research. No personally identifiable information was intentionally introduced into the synthetic training data.

7. ACKNOWLEDGMENTS

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